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### ML for the Discovery, Delivery, and Rapid Manufacturing of Potential Medicines

Dr. Pentelute opened by discussing deep learning to guide the discovery of potential antisense oligonucleotide treatments for genetic disease, using Duchenne muscular dystrophy as an example. The PMO oligomer was developed in 2016 as a treatment for Duchenne muscular dystrophy, working by skipping the DNA exon containing the mutation and allowing for the production of functional dystrophin. PMO is a large molecule, and its uptake by nuclei is limited. Because of this, the treatment has to be given in frequent, high doses. To combat this issue, PMO could be coupled with cell penetrating peptides to improve cellular uptake. The efficacy of these PMO-CPP complexes was determined by using fluorescent assays in HeLa EGFP654 cells. Changes in fluorescence were indicative of effective splicing of PMO into the cells DNA. It was determined that the most effective cell penetrating peptides were strings of arginine, which are toxic to cells at high enough doses. The lab created a library of data from the PMO-CPP complexes they studied to feed into deep learning models that were trained to optimize the cell penetrating peptides while limiting arginine content. They further refined these models using sequence and activity data from the CPPs. Additionally, they discovered that chimeric CPPs have a synergistic cell penetrating effect, beyond just being additive. They then “fingerprinted” building blocks of the peptides, including non-canonical amino acids. They used the deep learning model to predict slightly longer cell penetrating proteins that they called “Mach” mini proteins which included non-canonical amino acids. These mini proteins were more effective at delivering PMO into cells than the previously predicted CPPs. They determined the toxicity of the PMO-mini protein complexes using an LDH release assay in renal epithelial cells, and identified a dose-response relationship in heart, diaphragm, and quadricep tissues of mice. They also discovered that these mini proteins are effective at delivered other small molecules to cells. In the process of this research, the lab developed technology to rapidly and reliably synthesize peptides of over 100 residues. They produced the MDM2 protein, and the barnase enzyme through synthesis and recombinant expression; both methods produced nearly identical products and the two barnase products demonstrated comparable catalytic efficiency. Dr. Pentelute also founded a company dedicated to the rapid synthesis of peptides. The lab used techniques including affinity selection-mass spectrometry to identify ligands for a target compound, and create peptide encoded libraries that include canonical and non-canonical amino acids. Peptides are more chemically stable, and with the inclusion of non-canonical amino acids, there is potential for more diversity.

## References

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